

Notes: Altavilla, Ellul, Pagano, Polo, & Vlassopoulos (JFE, 2025)

Loan Guarantees, Bank Lending and Credit Risk Reallocation

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Institutional setting of the COVID guarantee programs	3
2.2	Credit-register and supervisory data	4
2.3	Sample structure	4
2.4	Key variables	4
2.5	Descriptive facts from Table 1 and Figures 2–5	5
3	Models and empirical specifications (all core equations + interpretation)	5
3.1	Project choice and the role of non-pledgeable returns	5
3.2	No-guarantee equilibrium	6
3.3	Competitive renegotiation after guarantees arrive	6
3.4	Bargaining power in renegotiation	7
3.5	Main bank-firm empirical specification	7
3.6	Firm-level empirical specification	7
3.7	Loan-term regressions	8
4	Identification and credibility	8
4.1	Core identification idea	8
4.2	Why this is credible	8
4.3	Selection of the guaranteed-loan bank	9
4.4	Robustness logic	9
4.5	What this paper is not	9
5	Tables and figures	9
5.1	Figure 1: Guaranteed loans and net lending across countries	9
5.2	Figures 2–4: Distribution of guaranteed lending across firms and countries	10

5.3	Table 1: Descriptive statistics	10
5.4	Figure 5: Distribution of firm-level credit substitution	11
5.5	Figure 6 and Table 2: Baseline within-firm substitution evidence	11
5.6	Table 3: Excluding relationships with maturing loans	12
5.7	Tables 4 and 5: Non-callable versus callable credit	12
5.8	Table 6: Which banks issue guaranteed loans?	14
5.9	Table 7: Including single-bank firms	14
5.10	Tables 8 and 9: Interest-rate concessions and the substitution tradeoff	14
5.11	Table 10: Maturity extension	15
5.12	Table 11: Firm-level substitution	15
6	Short summary	16
7	Conclusion	16
7.1	Core conclusion (what the paper establishes)	16
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** When governments guarantee bank loans, do banks expand total credit to firms, or do they partly use the guarantee to shift pre-existing credit risk from their own balance sheets onto the public sector?
- **Setting.** Euro-area COVID-19 guarantee programs, with the empirical analysis focused on the four largest countries: France, Germany, Italy, and Spain. The key period is February to August 2020, i.e., the first pandemic wave and the launch of guarantee programs.
- **Core empirical object.** The paper studies *credit substitution*: the fall in non-guaranteed lending by a bank after it grants a guaranteed loan to the same firm.
- **Main identification idea.** For firms borrowing from multiple banks, compare the lending behavior of the bank granting the guaranteed loan with that of other banks lending to the same firm. This is a within-firm design in the spirit of Khwaja and Mian (2008).
- **Main within-firm result.** One euro of guaranteed lending is associated with a reduction of about 0.28 euros in non-guaranteed credit by the bank offering the guaranteed loan, relative to other banks lending to the same firm.
- **Where substitution is strongest.** Substitution is larger for weaker firms — smaller, riskier, and more pandemic-exposed firms — and for stronger banks, especially those with fewer non-performing loans.
- **Loan-term result.** Banks granting guaranteed loans also cut interest rates and extend maturities relative to other lenders to the same firm. The paper interprets this as the other side of substitution: firms may lose some non-guaranteed quantity but gain better terms.
- **Firm-level result.** At the overall firm level, the effect is smaller: one euro of guaranteed lending is associated with only about 0.09 to 0.12 euros less non-guaranteed credit. Other banks partially offset the contraction by the guaranteed-loan bank.
- **Theory.** A simple renegotiation model predicts more substitution for riskier firms and a positive cross-sectional link between substitution and interest-rate reductions. The bank-firm distribution of bargaining power determines how much of the guarantee’s value accrues to the firm versus the bank.
- **Why the paper matters.** The paper shows that loan guarantees do not map one-for-one into new credit. They also reallocate credit risk from banks to taxpayers. This is a first-order policy issue when guarantees are used as crisis tools.

2 Data and measurement (key objects)

2.1 Institutional setting of the COVID guarantee programs

- **Common EU framework.** EU state-aid rules required that guaranteed loans go to firms that were *not in difficulty* as of December 31, 2019, but were adversely affected by the COVID shock afterwards.
- **Implication for selection.** Eligibility rules were designed to exclude both firms already close to insolvency before the pandemic and firms not negatively affected by COVID. So guarantees were not meant for the extreme lower or upper tails of firm risk.

- **Skin in the game.** The EU framework generally capped guarantee coverage at up to 90% under pari passu loss sharing, with some national exceptions and special programs reaching 100% for small loans.
- **Loan size and maturity rules.** Guaranteed amounts were tied to 2019 payroll or turnover, and the EU framework generally imposed a six-year maturity ceiling, though some national programs allowed longer loans.
- **Why this matters empirically.** Because guarantees were targeted and not randomly assigned, simple cross-firm comparisons would be contaminated by selection. This motivates the within-firm bank-level design.

2.2 Credit-register and supervisory data

- **Loan data.** The paper uses AnaCredit, the ECB’s granular euro-area credit register, at monthly frequency and loan level. It covers term loans, non-revolving credit lines, revolving credit, financial leases, and trade receivables.
- **Reporting threshold.** Company loans above 25,000 euros are reported under a common euro-area threshold, which helps cross-country comparability.
- **Guarantee identification.** AnaCredit records whether a loan is backed by a government guarantee, which lets the authors identify guaranteed lending during the pandemic.
- **Bank data.** Supervisory balance-sheet variables are drawn from ECB supervisory data and include bank size, liquidity coverage ratio, capital ratio, and the fraction of non-performing loans.

2.3 Sample structure

- **Universe before treatment.** The starting sample includes 2,527,108 firms in AnaCredit before the introduction of pandemic-related guarantee programs.
- **Country composition.** The sample includes 1,140,911 firms from France, 371,036 from Germany, 608,672 from Italy, and 406,489 from Spain.
- **Guaranteed-loan sample.** Between March and August 2020, 500,013 firms received guaranteed credit.
- **New-bank relationships.** About 13% of guaranteed-loan recipients obtained the guaranteed loan from a bank that had not been lending to them before the program.
- **Bank-firm sample for within-firm tests.** The main within-firm regressions use firms with multiple bank relationships, yielding 817,808 bank-firm observations in the baseline bank-firm panel.

2.4 Key variables

- **Guaranteed loan intensity.** At the bank-firm level, L_{ij}^G is the amount of guaranteed credit granted by bank j to firm i , divided by the firm’s total initial credit in February 2020.

- **Credit substitution.** The main dependent variable is the negative change in non-guaranteed credit from February to August 2020, scaled by initial total firm credit. A positive number means more substitution.
- **Firm fragility proxies.** The paper uses industry-country value-added growth during the pandemic, the log of firm debt as a proxy for size, and the fraction of loans in arrears as a proxy for ex-ante risk.
- **Bank strength proxies.** Bank size, liquidity, capital, and NPL ratios are all measured before the pandemic.
- **Loan terms.** Interest rates and residual maturities are observed at the loan level and aggregated to the bank-firm level when studying repricing and maturity changes.

2.5 Descriptive facts from Table 1 and Figures 2–5

- Firms receiving guaranteed loans are more exposed to pandemic-hit sectors but are *less* risky on average than firms in the full sample, consistent with program eligibility rules.
- The average guarantee share is about 0.81, the average guaranteed-loan interest rate is about 1.98%, and the average maturity is about 33 months.
- Spain and Italy account for the largest amounts of guaranteed lending, while Germany has the smallest aggregate amount but the largest average loan size.
- In every country, most guaranteed lending goes to SMEs, both by euro amount and, even more strongly, by number of firms.
- Firm-level substitution is positive for most firms that receive a guaranteed loan, with especially strong positive distributions in Italy and Spain.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Project choice and the role of non-pledgeable returns

Firms can run either a small project with cost I_s and success payoff Y , or a large project with cost $I_\ell > I_s$ and success payoff X . Large projects also yield a non-pledgeable private benefit B . Good firms prefer the large project if they could finance it internally:

$$B + (1 - p)(X - Y) > I_\ell - I_s. \quad (1)$$

Guaranteed credit has size L_G and is large enough to make the large project financeable for good firms but not bad ones:

$$(1 - p)X > I_\ell - L_G > (1 - p - \Delta)X. \quad (2)$$

- **Interpretation.** The guarantee relaxes credit constraints enough for good firms to scale up, but it is not meant to make lending to bad firms privately profitable on pledgeable cash flows alone.

- **Economic intuition.** Guarantees should matter most where banks want to shift default risk. That immediately suggests stronger substitution for riskier borrowers.

3.2 No-guarantee equilibrium

Without guarantees, only the small project is funded. Competitive lenders charge repayments

$$F_{sg} = \frac{I_s}{1-p}, \quad F_{sb} = \frac{I_s}{1-p-\Delta}. \quad (3)$$

Expected firm payoffs are

$$\Pi_{sg} = (1-p)Y - I_s, \quad \Pi_{sb} = (1-p-\Delta)Y - I_s. \quad (4)$$

- **Interpretation.** Bad firms pay a higher promised rate because their default probability is higher.
- **Baseline implication.** Before guarantees arrive, banks fully bear default risk on their loans.

3.3 Competitive renegotiation after guarantees arrive

If firms can renegotiate competitively, guaranteed lending displaces non-guaranteed lending on better terms. For firms that continue with the small project, promised repayments fall to

$$F'_{sg} = \frac{I_s - L_G}{1-p}, \quad F'_{sb} = \frac{I_s - L_G}{1-p-\Delta}. \quad (5)$$

Expected payoffs rise to

$$\Pi'_{sg} = \Pi_{sg} + pL_G, \quad \Pi'_{sb} = \Pi_{sb} + (p+\Delta)L_G. \quad (6)$$

Good firms may instead use the guarantee to move to the large project, in which case competitive repayment on the non-guaranteed part is

$$F_{\ell g} = \frac{I_\ell - L_G}{1-p}, \quad (7)$$

and expected payoff becomes

$$\Pi_{\ell g} = B + (1-p)X - I_\ell + pL_G > \Pi_{sg} + pL_G. \quad (8)$$

- **Main model prediction.** For bad firms, guaranteed loans lead to full substitution of non-guaranteed credit. For good firms, substitution is weaker and can even turn into net additional lending if the firm scales up into the larger project.
- **Rate prediction.** Because the guarantee shifts default risk to the government, the average cost of credit falls. The gain is larger for riskier firms because the guarantee is more valuable for them.

3.4 Bargaining power in renegotiation

If the initial bank has bargaining power, it can capture part of the guarantee's surplus. In the limiting case of a take-it-or-leave-it offer, the bank can force bad firms to promise

$$F''_{sb} = \frac{I_s}{1 - p - \Delta} - L_G, \quad (9)$$

which implies debt value

$$D''_{sb} = I_s + (p + \Delta)L_G. \quad (10)$$

For good firms financing the large project, the bank can similarly appropriate the expected value of the guarantee so that overall debt value becomes

$$D''_{\ell g} = I_\ell + pL_G. \quad (11)$$

- **Interpretation.** The same guarantee can produce different combinations of quantity adjustment and price adjustment depending on how much bargaining power the borrower has.
- **Cross-sectional implication.** Where firms have weaker bargaining power, guaranteed loans may still induce substitution, but banks pass through less of the guarantee via lower rates.

3.5 Main bank-firm empirical specification

The core within-firm regression is

$$\begin{aligned} s_{ij} = & \beta_1 L_{ij}^G + \beta_2 L_{ij}^G \times \Delta V A_i + \beta_3 L_{ij}^G \times Size_i + \beta_4 L_{ij}^G \times Risk_i \\ & + \beta_5 L_{ij}^G \times BSize_j + \beta_6 L_{ij}^G \times Liq_j + \beta_7 L_{ij}^G \times Cap_j + \beta_8 L_{ij}^G \times NPL_j \\ & + \gamma_i + \eta_j + \varepsilon_{ij}. \end{aligned} \quad (12)$$

where

$$s_{ij} = -\frac{NGC_{ij,t} - NGC_{ij,t-1}}{TC_{i,t-1}}. \quad (13)$$

- **Interpretation of s_{ij} .** Positive s_{ij} means bank j cut non-guaranteed credit to firm i between February and August 2020.
- **Interpretation of β_1 .** β_1 is the number of euros of non-guaranteed credit withdrawn per euro of guaranteed lending, relative to other banks lending to the same firm.
- **Why firm fixed effects matter.** γ_i means the comparison is within firm: guaranteed-loan banks are benchmarked against other lenders to the same borrower.

3.6 Firm-level empirical specification

At the firm level, substitution is measured by

$$s_i = -\frac{NGC_{i,t} - NGC_{i,t-1}}{TC_{i,t-1}}, \quad (14)$$

and estimated via

$$s_i = \alpha + \beta_1 L_i^G + \beta_2 L_i^G \times \Delta VA_i + \beta_3 L_i^G \times Size_i + \beta_4 L_i^G \times Risk_i \\ + \beta_5 L_i^G \times BSize_i + \beta_6 L_i^G \times Liq_i + \beta_7 L_i^G \times Cap_i + \beta_8 L_i^G \times NPL_i + \gamma_c + \varepsilon_i. \quad (15)$$

- **Interpretation.** This specification asks how total non-guaranteed credit to a firm changes when the firm receives more guaranteed credit.
- **Difference from the main design.** This is informative about total firm-level credit effects, but it is less cleanly identified because it does not use within-firm across-bank variation.

3.7 Loan-term regressions

The same bank-firm design is used with changes in average interest rates and residual maturities as dependent variables.

- **Interpretation.** The paper studies not just credit quantity but the full contract response: amount, price, and maturity.
- **Key prediction from the model.** More substitution should be associated with larger interest-rate concessions, especially where guarantees matter more for default risk.

4 Identification and credibility

4.1 Core identification idea

- **Problem.** Firms receiving guaranteed loans are not random. They are selected by formal eligibility rules and by banks' own screening.
- **Solution.** Restrict attention to firms with multiple bank relationships and compare the guaranteed-loan bank with the other banks lending to the same firm.
- **Counterfactual.** For a given firm, the lending response of non-guaranteed banks is the counterfactual for what the guaranteed-loan bank might have done absent the guarantee.
- **What is absorbed.** Firm fixed effects absorb all borrower-level pandemic exposure and pre-existing weakness common to that firm across lenders.

4.2 Why this is credible

- **Targeting does not kill identification.** The design does not rely on comparing guaranteed firms to non-guaranteed firms. It relies on comparing banks within the same guaranteed firm.
- **Bank fixed effects.** These absorb average differences in how aggressively banks lend during the pandemic.

- **Scaling choice.** The substitution variable is scaled by total initial firm credit rather than bank-level initial credit so that firms receiving guaranteed loans from new banks remain in the sample.

4.3 Selection of the guaranteed-loan bank

- **Empirical fact.** Within the same firm, the banks issuing guaranteed loans are larger, more capitalized, and more likely to be relationship lenders.
- **Bias direction.** These are banks that, if anything, should have supplied *more* credit during stress. Therefore the estimated substitution is likely a lower bound on the true effect for guaranteed-loan banks.
- **Residual-maturity evidence.** The guaranteed-loan bank tends to have longer pre-existing residual maturity, which works against the idea that the results are just expiring loans being mechanically replaced.

4.4 Robustness logic

- **Excluding maturing loans.** When relationships with loans maturing between March and August 2020 are removed, substitution estimates are almost unchanged.
- **Callable versus non-callable credit.** Substitution is much larger for revolving credit lines, which banks can revoke unilaterally, than for term loans and non-revolving lines.
- **Including single-bank firms.** Regressions with industry-location-size fixed effects and the full sample yield similar or slightly larger substitution estimates.

4.5 What this paper is not

- This is not an IV paper and does not claim identification from quasi-random policy discontinuities.
- The credibility comes from matched bank-firm microdata, within-firm comparisons, rich bank and relationship controls, and a battery of robustness checks tailored to the main threat: endogenous allocation of guarantees.

5 Tables and figures

5.1 Figure 1: Guaranteed loans and net lending across countries

Read:

- At the country level, net loan flows rise less than one-for-one with guaranteed-loan take-up during April–August 2020.
- France combines the largest guaranteed-loan take-up with the strongest positive net lending, while Spain has high take-up but a weaker net-lending response.
- The figure is only descriptive, but it motivates the paper’s central question: aggregate guaranteed lending does not mechanically translate into equal net new credit.

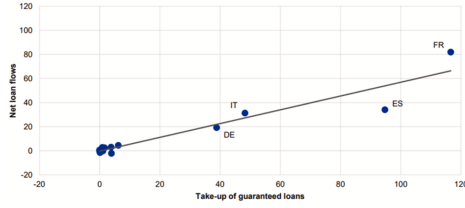


Fig. 1. Guaranteed loans and net lending: country-level data (billion euros)
This figure shows the cross-country relation between the amount of take-up of guaranteed loans and the net loan flows at country level, over the period April-August 2020. Each dot refers to a country in the euro area. Data sources: Kreditanstalt für Wiederaufbau for Germany, Instituto de Crédito Oficial for Spain, Ministère de l'Économie et des Finances for France, Ministero dell'Economia e delle Finanze and Banca d'Italia for Italy, various national authorities for other euro area countries, news sources, ECB and ECB calculations.

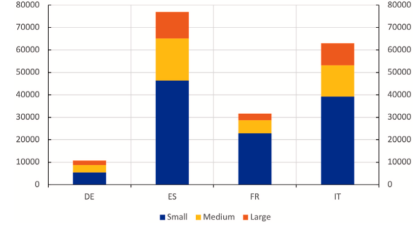


Fig. 2. Amount of guaranteed loans by firm size
The figure shows the amount of guaranteed loans (in million euros) issued to firms in different size classes based on their employment (small firms being those with less than 50 employees, medium firms those with 50 to 250 employees, and large firms as those with more than 250 employees). The sample includes firms present in the AnaCredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

5.2 Figures 2–4: Distribution of guaranteed lending across firms and countries

Read:

- Figures 2 and 3 show that SMEs dominate guaranteed-loan programs, both by euro amount and even more strongly by number of firms.
- Spain and Italy account for the largest total guaranteed amounts, while Germany has the smallest total amount.
- Figure 4 shows that German guaranteed loans are largest on average, Italian guaranteed loans smallest, and France and Spain sit in between.
- These figures show that program design and take-up differ across countries, which makes the cross-country robustness of the main results nontrivial.

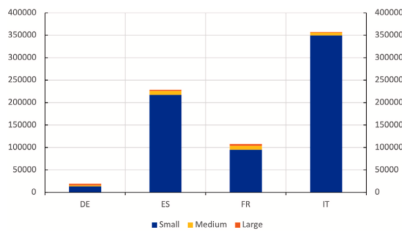


Fig. 3. Number of guaranteed loans by firm size
The figure shows the number of guaranteed loans issued to firms in different size classes to their employment size (small firms being those with less than 50 employees, medium firms those with 50 to 250 employees, large firms as those with more than 250 employees). The sample includes firms recorded in the AnaCredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

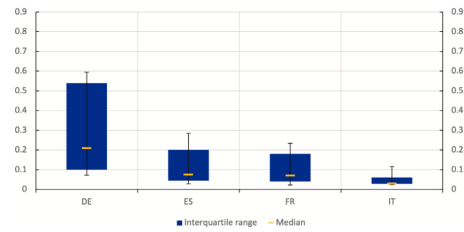


Fig. 4. Distribution of guaranteed loan size
The figure reports the distribution of the size of guaranteed loans in million euro issued in different countries. We report the median, the interquartile range and the 16th and 84th percentile. The sample includes firms recorded in the AnaCredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

5.3 Table 1: Descriptive statistics

Read:

- The full sample has more than 2.5 million firms, while about 500,000 firms receive guaranteed credit.
- Firms receiving guaranteed loans are in more pandemic-hit sectors but are less risky ex ante than the average firm, consistent with eligibility screening.
- For guaranteed loans, the average guarantee share is about 81%, the average rate is about 1.98%, and average maturity is 33 months.
- The bank-firm sample used in the main within-firm design contains 817,808 observations for firms with multiple banking relationships that receive guaranteed credit.

Table 1
Descriptive statistics.

Variable	Observations	Mean	Std. dev.	p25	p50	p75
Panel (a): Firm-level statistics for full sample						
G_i	2,527,108	0.22	0.41	0.00	0.00	0.00
L_i^G	2,527,108	0.15	0.40	0.00	0.00	0.00
Industry VA Growth	2,527,108	-0.18	0.13	-0.28	-0.21	-0.04
Firm Size (ln)	2,527,108	-1.84	1.39	-2.80	-1.96	-1.02
Firm Risk	2,527,108	0.05	0.20	0.00	0.00	0.00
Bank Assets (ln)	2,492,992	11.02	1.87	9.75	10.68	12.70
Bank Liquidity	1,970,977	1.59	0.44	1.37	1.45	1.64
Bank Capital	2,461,231	0.17	0.04	0.14	0.16	0.19
Bank NPL	2,478,597	0.04	0.02	0.02	0.03	0.04
Variable	Observations	Mean	Std. Dev.	p25	p50	p75
Panel (b): Firm-level statistics for sample of firms receiving a guaranteed loan						
s_i	500,013	0.25	0.53	0.01	0.20	0.59
L_i^G	500,013	0.71	0.61	0.26	0.55	0.97
Guar. Amount	474,103	0.59	0.53	0.21	0.44	0.80
Industry VA Growth	500,013	-0.27	0.12	-0.30	-0.28	-0.25
Firm Size (ln)	500,013	-1.71	1.44	-2.77	-1.87	-0.79
Firm Risk	500,013	0.02	0.10	0.00	0.00	0.00
Bank assets (ln)	495,051	11.66	1.61	10.51	12.09	12.92
Bank Liquidity	462,323	1.63	0.40	1.37	1.52	1.72
Bank Capital	488,049	0.16	0.03	0.14	0.16	0.18
Bank NPL	493,273	0.05	0.03	0.03	0.04	0.07

Panel (c): Bank-firm-level statistics for sample of firms with multiple bank relationships receiving a guaranteed loan

s_{ij}	817,808	0.04	0.30	0.00	0.01	0.11
G_{ij}	817,808	0.51	0.50	0.00	1.00	1.00
L_{ij}^G	817,808	0.20	0.35	0.00	0.00	0.25
Industry VA Growth	813,774	-0.28	0.12	-0.30	-0.27	-0.25
Firm Size (ln)	806,194	-0.42	1.56	-1.56	-0.53	0.62
Firm Risk	812,803	0.02	0.09	0.00	0.00	0.00
Bank assets (ln)	813,781	11.32	1.76	10.10	11.74	12.92
Bank Liquidity	670,763	1.70	0.48	1.32	1.54	1.79
Bank Capital	791,748	0.15	0.03	0.14	0.14	0.17
Bank NPL	803,958	0.05	0.03	0.03	0.04	0.06
Share of gran.	817,808	0.32	0.24	0.12	0.27	0.48
Drawn/Granted	808,512	0.83	0.29	0.71	0.98	1.00
Residual Maturity (ln)	737,745	3.11	1.42	2.60	3.53	3.98
Residual Maturity	816,112	34.61	8.07	30.90	51.03	
Panel (d): Loan-level statistics for guaranteed loans						
Guar. Percent	580,775	0.81	0.21	0.80	0.80	1.00
Interest Rate	995,288	1.98	1.34	1.21	1.65	2.63
Maturity	578,893	33	26.20	26	36	60

This table reports the descriptive statistics for the variables used in our analysis. Panel (a) reports the statistics at the firm level for the full sample, Panel (b) reports them for the sample of firms receiving a guaranteed loan, Panel (c) reports bank-firm-level statistics for the sample of firms with multiple bank relationships receiving a guaranteed loan, and Panel (d) reports loan-level statistics for guaranteed loans. The dummy G_i equal to 1 if firm i receives a government guaranteed loan between March 2020 and August 2020, and 0 otherwise (at the bank-firm level, the dummy G_{ij} is equal to 1 if bank j gives a government guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise); L_i^G is measured at the firm level and is defined as the amount of the government guaranteed loan received by the firm, divided by total credit in February 2020, while L_{ij}^G is defined at the bank-firm level and is defined as the amount of the government guaranteed loan received by the firm i by bank j , divided by total initial credit granted to firm i in February 2020; Industry VA Growth is defined as the percentage change in Value Added in the relevant industrial sector in each country between February 2020 and August 2020; Firm Size is proxied by the log of firm total debt; Firm Risk is proxied by the share of loans in arrears out of total loans; Bank Assets is defined as the log of total bank assets; Bank Liquidity is defined as the bank Liquidity Coverage Ratio; Bank Capital is defined as the Core Tier 1 Ratio; Bank NPL is defined as the share of NPL loans out of its total loans; Credit substitution (s_i) is measured at firm level and is defined as the negative of the change in non-guaranteed credit received by firm i between February 2020 and August 2020, divided by its total credit as of February 2020, while s_{ij} is measured at the bank-firm level and is defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020; Guar. Amount is defined as the amount of the government-guaranteed credit multiplied by the percentage of guarantee received by the firm, divided by total credit in February 2020; Share of granted is defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted is defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i ; Guar. Percent is the percentage of the guaranteed loan covered by the public guarantee. Interest Rate is the rate charged on the guaranteed loan; Maturity is the log of the number of months of the guaranteed loan's maturity. In Panels (a) and (b) bank-level variables are measured as weighted averages of the values for the banks lending to a given firm, where the weights are the shares of the bank exposure toward the firm in the firm's total bank exposure as of December 2019. All variables, apart from Guarantee, Industry VA Growth, Interest rate, and Maturity are measured as of December 2019.

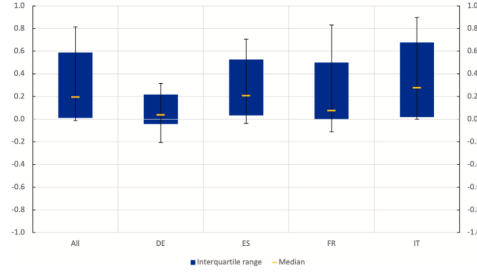


Fig. 5. Distribution of firm-level credit substitution
The figure shows the country-level distribution of the firm-level credit substitution defined as the negative of the change in non-guaranteed credit between February 2020 and August 2020, divided by total credit in February 2020 (s_i for firm i). Each box plot displays the median, the interquartile range and the 16th and 84th percentile. The sample includes all firms in the AnaCredit database as of December 2019 that received a guaranteed loan between March and August 2020.

5.4 Figure 5: Distribution of firm-level credit substitution

Read:

- The firm-level substitution measure is positive for most firms receiving guaranteed credit.
- Medians are positive in all countries and especially strong in Italy and Spain.
- Even before controls, the picture is that guaranteed lending often comes with some reduction in other non-guaranteed credit.

5.5 Figure 6 and Table 2: Baseline within-firm substitution evidence

Read:

- Figure 6 shows that banks granting guaranteed loans reduce non-guaranteed exposures, while other banks lending to the same firm on average expand theirs.
- In the full multiple-bank sample, guaranteed-loan banks reduce non-guaranteed credit by about 12% on average, whereas non-guaranteed banks increase exposure by about 5%.
- Table 2 translates this into regression magnitudes: in the simplest specification, one euro

of guaranteed lending reduces non-guaranteed lending by 0.291 euros; in the most complete specification the estimate is 0.281.

- Substitution is larger for firms in more affected sectors, for smaller firms, and for riskier firms. It is also larger for stronger banks, especially banks with fewer NPLs.

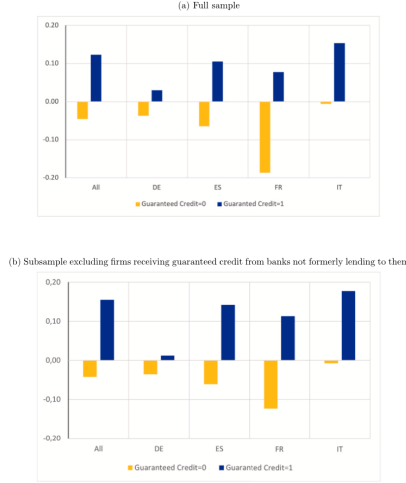


Fig. 6. Credit substitution for bank-firm relationships with and without guaranteed credit
The figure reports bank-firm level descriptive statistics of the variable s_{ij} , defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The figure shows the average value of s_{ij} for bank-firm relationships where bank j issues a government-guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise. Panel (a) is based on the sample of all firms in the Anascredit database as of December 2019 that received a government-guaranteed loan and had multiple bank relationships. Panel (b) is based on a subsample that excludes firms whose guaranteed credit was issued by new banks, i.e., banks not lending to them as of February 2020.

Table 2
Credit substitution: firm-bank level analysis.

Dependent variable:	Credit Substitution (s_{ij})				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.291*** (0.016)	0.322*** (0.018)	0.278*** (0.016)	0.278*** (0.017)	0.281*** (0.016)
$L_{ij}^G \times \text{Industry VA Growth}$		-0.190** (0.076)	-0.089* (0.051)	-0.089* (0.048)	-0.118** (0.049)
$L_{ij}^G \times \text{Firm Size}$		-0.036*** (0.011)	-0.033*** (0.010)	-0.033*** (0.010)	-0.028*** (0.010)
$L_{ij}^G \times \text{Firm Risk}$		0.341*** (0.062)	0.324*** (0.059)	0.311*** (0.060)	0.132** (0.057)
$L_{ij}^G \times \text{Bank Assets}$		0.013** (0.007)	0.002 (0.007)	0.004 (0.008)	0.009 (0.007)
$L_{ij}^G \times \text{Bank Liquidity}$		0.001*** (0.000)	0.000* (0.000)	0.003*** (0.000)	0.001*** (0.000)
$L_{ij}^G \times \text{Bank Capital}$		0.891* (0.527)		1.046** (0.478)	0.927** (0.417)
$L_{ij}^G \times \text{Bank NPL}$			-2.757*** (0.510)	-2.953*** (0.540)	-3.865*** (0.587)
$L_{ij}^G \times \text{Share of Granted}$					-0.518*** (0.032)
$L_{ij}^G \times \text{Drawn/Granted}$					-0.229*** (0.044)
$L_{ij}^G \times \text{Residual Maturity}$					0.046** (0.020)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R ²	0.502	0.495	0.497	0.498	0.532
N	817 808	599 302	599 360	589 477	509 100

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij} , defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Value Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.6 Table 3: Excluding relationships with maturing loans

Read:

- Once bank-firm relationships with loans maturing during March–August 2020 are excluded, the baseline substitution estimate remains around 0.26 to 0.31.
- The most complete specification yields 0.262, very close to the main benchmark.
- This table is important because it shows the results are not mechanically driven by expiring non-guaranteed loans being replaced by guaranteed loans.

5.7 Tables 4 and 5: Non-callable versus callable credit

Read:

- For term loans and non-revolving credit lines, the most complete substitution estimate is about 0.252.
- For revolving credit lines, which the bank can revoke unilaterally, the most complete estimate is about 0.427.
- The economic message is clear: banks substitute away much more from credit instruments they can adjust without renegotiation.

Table 3
Credit substitution: firm-bank level analysis, excluding relationships with maturing loans.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.281*** (0.015)	0.308*** (0.021)	0.269*** (0.019)	0.259*** (0.020)	0.262*** (0.019)
L_{ij}^G xIndustry VA Growth		-0.057 (0.050)	-0.001 (0.044)	-0.016 (0.045)	-0.020 (0.036)
L_{ij}^G xFirm Size		-0.034*** (0.009)	-0.032*** (0.009)	-0.032*** (0.010)	-0.028*** (0.010)
L_{ij}^G xFirm Risk		0.336*** (0.060)	0.309*** (0.059)	0.302*** (0.060)	0.132** (0.063)
L_{ij}^G xBank Assets		0.005 (0.010)	0.004 (0.008)	0.014 (0.011)	0.015** (0.008)
L_{ij}^G xBank Liquidity		0.016 (0.048)	0.022 (0.044)	0.017 (0.046)	0.006 (0.035)
L_{ij}^G xBank Capital		1.045** (0.480)	0.980* (0.511)	0.830** (0.405)	0.830** (0.554)
L_{ij}^G xBank NPL			-2.773*** (0.482)	-3.001*** (0.506)	-3.872*** (0.554)
L_{ij}^G xShare of Granted					-0.511*** (0.033)
L_{ij}^G xDrawn/Granted					-0.241*** (0.044)
L_{ij}^G xResidual Maturity					0.045** (0.021)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R ²	0.503	0.494	0.497	0.498	0.531
N	705 965	510 099	500 477	491 345	417 784

This table reports bank-firm level estimates of a regression whose dependent variable is the credit substitution $s_{i,j}$, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The sample excludes bank-firm relationships where there is at least one loan maturing between March 2020 and August 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm divided by the total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 5
Credit substitution: firm-bank level analysis. Revolving credit lines.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.426*** (0.082)	0.464*** (0.093)	0.402*** (0.094)	0.424*** (0.106)	0.427*** (0.115)
L_{ij}^G xIndustry VA Growth		-0.808*** (0.249)	-0.689*** (0.231)	-0.634*** (0.222)	-0.367* (0.188)
L_{ij}^G xFirm Size		-0.089*** (0.017)	-0.091*** (0.015)	-0.088*** (0.017)	-0.125*** (0.030)
L_{ij}^G xFirm Risk		0.777*** (0.219)	0.830*** (0.216)	0.840*** (0.217)	1.790*** (0.350)
L_{ij}^G xBank Assets		0.142*** (0.028)	0.0942*** (0.030)	0.112*** (0.031)	0.111** (0.043)
L_{ij}^G xBank Liquidity		0.552*** (0.133)	0.407*** (0.124)	0.513*** (0.132)	0.471*** (0.161)
L_{ij}^G xBank Capital		4.407*** (1.085)		3.930*** (1.017)	2.955*** (1.081)
L_{ij}^G xBank NPL			-3.363*** (1.853)	-3.862*** (1.863)	-3.386*** (2.309)
L_{ij}^G xShare of gran.					-0.574*** (0.212)
L_{ij}^G xDrawn/Granted					-0.609*** (0.222)
L_{ij}^G xResidual Maturity					0.151*** (0.071)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
N	428 761	326 285	327 720	323 854	256 201
r2	0.669	0.752	0.752	0.753	0.624

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution $s_{i,j}$, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. In these regressions, "credit" refers exclusively to revolving credit lines. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 4
Credit substitution: firm-bank level analysis. Term loans plus non-revolving credit lines.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.138*** (0.018)	0.249*** (0.044)	0.212** (0.073)	0.266** (0.056)	0.252** (0.093)
L_{ij}^G xIndustry VA Growth		-1.180* (0.615)	-0.695* (0.386)	-0.533 (0.436)	-0.546 (0.422)
L_{ij}^G xFirm Size		-0.241*** (0.082)	-0.147** (0.069)	-0.141* (0.073)	-0.136* (0.075)
L_{ij}^G xFirm Risk		3.415*** (0.620)	3.044*** (0.511)	3.083*** (0.510)	3.503*** (0.615)
L_{ij}^G xBank Assets		0.412*** (0.133)	0.501*** (0.118)	0.495*** (0.115)	0.391*** (0.116)
L_{ij}^G xBank Liquidity		0.922** (0.437)	1.143*** (0.329)	0.995*** (0.323)	0.726** (0.311)
L_{ij}^G xBank Capital		5.540** (5.630)		3.309 (2.429)	2.938 (2.830)
L_{ij}^G xBank NPL			-4.601*** (0.432)	-4.855*** (0.413)	-5.101*** (0.436)
L_{ij}^G xShare of gran.					-2.647*** (0.306)
L_{ij}^G xDrawn/Granted					-0.807*** (0.209)
L_{ij}^G xResidual Maturity					0.021 (0.103)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
N	335 266	239 588	241 806	237 284	213 349
r2	0.678	0.716	0.718	0.720	0.545

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution $s_{i,j}$, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. In these regressions, "credit" refers exclusively to term loans plus non-revolving credit lines. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 6
Characteristics of banks providing guaranteed loans.

Dependent variable:	Guaranteed loan ($G_{i,j}$)					
	(1)	(2)	(3)	(4)	(5)	(6)
Bank Assets	0.055*** (0.001)	0.071*** (0.023)	0.059*** (0.013)	0.069*** (0.012)	0.086*** (0.026)	0.071*** (0.014)
Bank Liquidity	0.048 (0.282)	0.061 (0.040)	0.011 (0.030)	0.002 (0.030)	0.065 (0.043)	0.005 (0.033)
Bank Capital	3.704*** (0.616)		3.687*** (0.596)	4.281*** (0.784)	4.257*** (0.762)	
Bank NPL		-1.215 (1.620)	-0.788 (1.372)		-1.090 (1.953)	-0.261 (1.677)
Share of Granted	0.648*** (0.036)	0.676*** (0.036)	0.651*** (0.037)	0.430*** (0.038)	0.466*** (0.049)	0.430*** (0.038)
Drawn/Granted	0.206*** (0.075)	0.167*** (0.062)	0.207*** (0.075)	0.0750 (0.059)	0.107** (0.054)	0.149** (0.059)
Residual Maturity (ln)				0.109*** (0.021)	0.103*** (0.026)	0.110*** (0.021)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.401	0.367	0.405	0.441	0.397	0.445
N	629 964	630 117	619 623	554 537	555 058	545 767

The table reports bank-firm level estimates of a regression whose dependent variable is a dummy $G_{i,j}$ equal to 1 if bank j gives a government-guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise. The explanatory variables are bank and firm characteristics. The bank variables are: Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of total loans. The bank-firm variables are: Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables are measured as of December 2019. Standard errors, clustered at the bank level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

5.8 Table 6: Which banks issue guaranteed loans?

Read:

- Within the same firm, the bank providing the guaranteed loan is more likely to be larger, better capitalized, and a relationship lender with a larger share of the firm's credit.
- Residual maturity is also higher for the guaranteed-loan bank, which pushes against the “mechanical rollover” story.
- Because these selected banks are precisely the kinds of banks that usually lend more in bad times, the main substitution estimates likely understate the true underlying substitution tendency.

5.9 Table 7: Including single-bank firms

Read:

- When the sample is expanded to include single-bank firms and the regression uses industry-location-size fixed effects instead of firm fixed effects, results remain similar.
- The most complete estimate is 0.307, slightly larger than the within-firm benchmark.
- This shows the main message is not an artifact of restricting the sample to firms with multiple bank relationships.

Table 7
Substitution: firm-bank level analysis, including single-bank firms.

Dependent variable:	Credit substitution (s_{ij})				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.337*** (0.028)	0.342*** (0.025)	0.343*** (0.019)	0.326*** (0.021)	0.307*** (0.019)
L_{ij}^G x Industry VA Growth		0.141 (0.103)	0.114 (0.103)	0.103 (0.082)	0.081* (0.046)
L_{ij}^G x Firm Size		-0.009 (0.007)	-0.007 (0.007)	-0.009 (0.007)	-0.026*** (0.008)
L_{ij}^G x Firm Risk		0.493*** (0.052)	0.463*** (0.043)	0.456*** (0.046)	0.453*** (0.048)
L_{ij}^G x Bank Assets		0.010 (0.015)	0.017 (0.015)	0.002 (0.012)	0.000 (0.010)
L_{ij}^G x Bank Liquidity		0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.001** (0.000)
L_{ij}^G x Bank Capital		1.080 (0.781)	1.613** (0.685)	1.258** (0.563)	1.258** (0.563)
L_{ij}^G x Bank NPL			-2.450*** (0.676)	-3.138*** (0.828)	-2.169*** (0.322***)
L_{ij}^G x Share of Granted					-0.322*** (0.035)
L_{ij}^G x Drawn/Granted					-0.079 (0.117)
L_{ij}^G x Residual Maturity					0.100*** (0.033)
Industry-Location-Size FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R ²	0.130	0.126	0.127	0.130	0.157
N	3561108	2555452	2563497	2514094	2102075

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij} , defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 8
Firm-bank level analysis of changes in interest rates.

Dependent variable:	$\Delta InterestRate_{ij}$					
	(1)	(2)	(3)	(4)	(5)	(6)
L_{ij}^G	-0.387*** (0.041)	-0.449*** (0.074)	-0.488*** (0.073)	-0.515*** (0.079)	-0.559*** (0.070)	-0.426*** (0.066)
L_{ij}^G x Industry VA Growth		0.466*** (0.127)	0.397*** (0.121)	0.392*** (0.120)	0.452*** (0.118)	0.181*** (0.060)
L_{ij}^G x Firm Size		0.072*** (0.016)	0.076*** (0.017)	0.076*** (0.016)	0.066*** (0.016)	0.060*** (0.016)
L_{ij}^G x Firm Risk		0.149 (0.442)	0.118 (0.431)	0.103 (0.449)	-1.513** (0.674)	-0.382** (0.181)
L_{ij}^G x Bank Assets		0.043 (0.028)	0.027 (0.026)	0.026 (0.029)	0.035 (0.029)	0.027 (0.021)
L_{ij}^G x Bank Liquidity		0.034 (0.127)	0.033 (0.126)	0.038 (0.129)	0.034 (0.133)	0.024 (0.132)
L_{ij}^G x Bank Capital		0.119 (0.915)	0.838 (0.980)	0.463 (1.066)	1.746*** (1.429)	0.604 (0.953)
L_{ij}^G x Bank NPL			-3.145*** (1.209)	-3.172*** (1.336)	-4.399*** (1.179)	-2.871*** (0.773)
L_{ij}^G x Share of Granted					-1.170*** (0.139)	-0.773*** (0.134)
L_{ij}^G x Drawn/Granted					0.585*** (0.083)	0.457*** (0.052)
L_{ij}^G x Residual Maturity					-0.152* (0.084)	-0.040 (0.066)
L_{ij}^G x Lending Rate						-37.480*** (2.160)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes	Yes
R ²	0.388	0.402	0.402	0.403	0.412	0.728
N	767159	557527	557383	547697	474338	469215

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta InterestRate_{ij}$, charged by bank j to firm i between February 2020 and August 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

5.10 Tables 8 and 9: Interest-rate concessions and the substitution tradeoff

Read:

- Table 8 shows that banks granting guaranteed loans cut average interest rates relative to other lenders to the same firm. The paper interprets the average effect as about 17 basis points.

- Rate cuts are larger for smaller, riskier, and more pandemic-exposed firms, which matches the model's prediction that guarantees are more valuable for fragile borrowers.
- Table 9 shows a direct tradeoff between quantity and price: more credit substitution is associated with larger interest-rate reductions. In the full sample, 100% substitution corresponds to about a 79-bp rate cut.
- This tradeoff depends on bargaining power. For sole-bank firms, the implied cut is only about 28 bp, versus about 109 bp for multi-bank firms. Firms with revolving credit also receive smaller concessions than firms without revolving credit.

Table 9
Reduction in interest rate and credit substitution.

Dependent variable:	$\Delta InterestRate_{ij}$				
	Full Sample	Firms borrowing from one bank	Firms borrowing from multiple banks	Firms with revolving credit	Firms without revolving credit
	(1)	(2)	(3)	(4)	(5)
s_{ij}	-0.788*** (0.100)	-0.281*** (0.0718)	-1.090*** (0.146)	-0.596*** (0.106)	-0.738*** (0.138)
Bank FE	Yes	Yes	Yes	Yes	Yes
R ²	0.0539	0.0773	0.0493	0.0401	0.198
N	680 665	258 997	421 591	488 906	191 624

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta InterestRate_{ij}$ charged by bank j to firm i between February 2020 and August 2020. The explanatory variable is credit substitution s_{ij} , defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The sample is made by only those bank-firm relationships with a guaranteed loan between February 2020 and August 2020. We include both firms with single and multiple banks. In column 1 we consider the full sample. In columns 2 and 3 we split the sample between firms borrowing only from one bank (column 2) and firms borrowing from multiple banks (column 3). In columns 4 and 5 we split the sample between firms with at least one revolving credit instrument in February 2020 (column 4) and firms without revolving credit instruments in February 2020. Robust standard errors are in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10
Firm-bank level analysis of changes in loan maturities.

Dependent variable:	$\Delta Maturity_{ij}$				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	23.612*** (2.031)	25.754*** (1.556)	21.681*** (2.176)	21.513*** (1.931)	17.532*** (2.325)
$L_{ij}^G \times \text{Industry VA Growth}$		-4.265 (5.058)	-2.081 (5.555)	-2.078 (5.406)	-3.358 (3.937)
$L_{ij}^G \times \text{Firm Size}$		-0.255 (0.414)	-0.104 (0.429)	-0.0315 (0.436)	-0.139 (0.560)
$L_{ij}^G \times \text{Firm Risk}$		16.742*** (3.952)	19.401*** (4.368)	19.374*** (4.514)	5.946 (4.945)
$L_{ij}^G \times \text{Bank Assets}$		2.030*** (0.598)	0.300 (0.796)	0.459 (0.721)	0.817 (0.819)
$L_{ij}^G \times \text{Bank Liquidity}$		0.018 (0.017)	0.037 (0.024)	0.037* (0.021)	0.044** (0.020)
$L_{ij}^G \times \text{Bank Capital}$		-146.336*** (25.961)		-135.928*** (28.433)	-67.954*** (19.752)
$L_{ij}^G \times \text{Bank NPL}$			347.172*** (59.642)	327.514*** (56.341)	201.025*** (45.061)
$L_{ij}^G \times \text{Share of Granted}$					-54.382*** (5.272)
$L_{ij}^G \times \text{Drawn/Granted}$					4.058* (2.202)
$L_{ij}^G \times \text{Residual Maturity}$					8.129*** (1.115)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R ²	0.348	0.371	0.371	0.371	0.406
N	814 348	597 034	597 205	587 343	507 143

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta Maturity_{ij}$ in the bank-firm relationship between bank j and firm i between February 2020 and August 2020. The main explanatory variable, L_{ij}^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Value Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i ; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i ; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i . All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.11 Table 10: Maturity extension

Read:

- Banks granting guaranteed loans also lengthen maturities relative to other banks lending to the same firm.
- The paper summarizes the average effect as about 9 additional months for the average guaranteed relationship.
- The effect is stronger for riskier firms, but relationship features matter: banks with a larger existing share of the firm's credit extend maturities less, while relationships with longer pre-existing residual maturity see larger extensions.

5.12 Table 11: Firm-level substitution

Read:

- At the firm level, guaranteed lending still crowds out some non-guaranteed credit, but much less than at the guaranteed-bank level.

Table 11
Credit substitution: firm-level analysis.

Dependent variable:	Credit substitution (s_i)				
	(1)	(2)	(3)	(4)	5
L_i^G	0.107*** (0.021)	0.120*** (0.026)	0.113*** (0.020)	0.095*** (0.013)	0.086*** (0.013)
L_i^G x Industry VA Growth		-0.041 (0.060)	-0.052 (0.052)	-0.013 (0.055)	-0.018 (0.037)
L_i^G x Firm Size		-0.023*** (0.006)	-0.025*** (0.007)	-0.020*** (0.006)	-0.022*** (0.006)
L_i^G x Firm Risk		0.340*** (0.057)	0.352*** (0.061)	0.328*** (0.052)	0.318*** (0.057)
L_i^G x Bank assets			0.001 (0.012)	0.006 (0.012)	0.010 (0.010)
L_i^G x Bank Liquidity			0.001*** (0.000)	0.000** (0.000)	0.002*** (0.000)
L_i^G x Bank Capital			1.222* (0.625)		1.639*** (0.511)
L_i^G x Bank NPL				-2.870*** (0.487)	-3.352*** (0.550)
Country FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	Yes	Yes	Yes	Yes	Yes
R ²	0.034	0.040	0.053	0.046	0.060
N	623 133	500 013	457 697	460 275	456 306

The table reports firm-level estimates of a regression whose dependent variable is the credit substitution s_i , defined as the negative of the change in non-guaranteed credit received by firm i between February 2020 and August 2020, divided by its total credit as of February 2020. The main explanatory variable, L_i^G , is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Value Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans. Each of the bank related variables is calculated as a weighted average of the corresponding bank-level variable, where the weights are the shares of the banks' exposure toward the firm out of total bank exposure of the firm at December 2019. All the explanatory variables, apart from L_i^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the main bank level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

- Depending on the specification, one euro of guaranteed lending is associated with roughly 0.086 to 0.120 euros less non-guaranteed credit.
- The most complete specification gives 0.086, while the text emphasizes a range of roughly 9 to 12 cents.
- Smaller, riskier firms and firms borrowing from stronger banks again show more substitution.
- The contrast with the within-firm estimate means other lenders partially offset the guaranteed bank's retrenchment.

6 Short summary

- **Conclusion.** Government-guaranteed lending during COVID did expand firm credit, but not one-for-one: banks partly substituted guaranteed credit for their own non-guaranteed exposure.
- **Intuition.** Guarantees insure default risk. Banks therefore have an incentive to reallocate credit risk to the public sector, especially for weaker borrowers. Firms partly accept this because guarantees also buy lower rates and longer maturities.
- **Empirical lesson.** To evaluate guarantee programs, one must look at net credit and contract terms, not just gross guaranteed lending.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper shows that euro-area COVID guarantee programs generated economically meaningful credit-risk reallocation from banks to the public sector.
- **Baseline magnitude.** The preferred within-firm estimate is that one euro of guaranteed lending displaced about 28 cents of non-guaranteed credit by the guaranteed-loan bank relative to other banks lending to the same firm.

- **Firm-level net effect.** Because other lenders often expand credit, the net firm-level displacement is much smaller, around 9 to 12 cents per euro of guaranteed lending.
- **Cross-sectional pattern.** Substitution is strongest for smaller, riskier, and more pandemic-exposed firms, and for stronger banks.
- **Loan-term pattern.** Banks do not only change quantities. They also lower rates and extend maturities, especially for fragile firms. That is the contract-level mirror image of substitution.
- **Credibility.** The results survive excluding maturing loans, splitting by callable versus non-callable credit, and expanding the sample to include single-bank firms.

7.2 Economic intuition

- Loan guarantees create an insured funding channel. Once that channel opens, a bank can preserve the relationship with the firm while shifting part of the risk to the government.
- This incentive is strongest when the borrower’s default risk is high, because the value of public insurance is then especially large.
- But the firm is not necessarily worse off in every dimension. The guarantee gives the bank room to cut rates and lengthen maturities, so the final contract is a mix of quantity retrenchment and price relief.
- Bargaining power determines how that mix is split. Firms with more outside options, such as firms borrowing from multiple banks, capture more of the guarantee through lower rates.

7.3 Takeaway

This paper’s central lesson is that guaranteed-loan programs should be evaluated as tools for both credit expansion and credit-risk transfer. During the COVID crisis, euro-area guarantees did keep credit flowing, but they also allowed banks to partially de-risk their balance sheets. The right policy interpretation is therefore nuanced: the gross volume of guaranteed loans overstates the amount of genuinely additional credit delivered to firms, yet that does not automatically imply policy failure. Absent guarantees, banks might have cut exposures even more aggressively; by moving some risk to the public sector, banks may also have preserved lending capacity for the recovery; and by tempering credit growth to the riskiest firms, substitution may have limited post-crisis leverage overhang. The paper’s real contribution is to force the reader to evaluate guarantee programs on all three margins — quantity, price, and risk transfer — rather than on gross guaranteed lending alone.